DIGITAL BATTERY PASSPORT

DBP-NASA-B0007

Battery B0007 · Status repurposed · Second-life grade B

VERIFIED: all 18 events pass the six checks

Access tier: legitimate interest (full). Schema aligned with Regulation (EU) 2023/1542, Article 77 and Annex XIII.

Scan to open the live record and check that it matches this copy.

General information

model	B0007
manufacturer	Unknown (NASA Ames test cells)
manufacture date	2008-03-03
chemistry	Li-ion (LiCoO2-based 18650)
form factor	18650 cylindrical
rated capacity ah	2
nominal voltage v	3.7
manufacturing site	not disclosed (laboratory test cell)

Carbon footprint

kgco2e per kwh	75
total kgco2e	0.555
basis	assumed mid value of published 61-106 kg CO2e/kWh (IVL)

Performance and durability

rated capacity ah	2
initial capacity ah	1.881
initial resistance ohm	0.0997
expected cycle life	not recorded
cut off voltage v	not recorded

Battery status

status	repurposed
events	18
last event	SOH_ASSESSMENT

State of health (restricted)

assessment: cycle	150
assessment: soh	0.6923
assessment: soh lo	0.6326
assessment: soh hi	0.752
assessment: soh measured	0.7209
assessment: resistance growth	1.539
assessment: method	Gradient Boosting on charge-phase signals, 95% conformal interval
rul: cycle	80
rul: threshold	0.8
rul: rul cycles	6.4
rul: rul lo	0

rul: rul hi	22.3
rul: method	quantile gradient boosting, conformalised (80% band)

Lifecycle decision (restricted)

cycle	90
grade	B
application	Grid / home energy storage
confidence	high
recommendation	Repurpose for Grid / home energy storage
service years	not recorded
co2 avoided kg per 60kwh pack	3658

Signed event chain (18 events)

#	Date	Event	Signed by	Hash
0	2008-03-03	Manufactured	MFR-01	`60ee3beb626dd06f2f3d…`
1	2008-03-04	Carbon Footprint Declared	MFR-01	`39586695c2a8e1604810…`
2	2008-04-02	First Life Start	OEM-01	`63cf1b3bd8e9a5ffb6a2…`
3	2008-04-18	Soh Assessment	AI-ASSESSOR	`1e012af0cd745b5ed796…`
4	2008-04-18	Rul Prediction	AI-ASSESSOR	`a79b2ec02e87bf3b3637…`
5	2008-04-24	Soh Assessment	AI-ASSESSOR	`47e81145e61dc493c8c8…`
6	2008-04-24	Rul Prediction	AI-ASSESSOR	`860688a4611ab2de19b4…`
7	2008-05-01	Soh Assessment	AI-ASSESSOR	`05d62180e74fe569760e…`
8	2008-05-01	Rul Prediction	AI-ASSESSOR	`fed899ceadd68d4d3bc9…`
9	2008-05-06	Soh Assessment	AI-ASSESSOR	`6febeacfaa4d13ab325b…`
10	2008-05-06	Rul Prediction	AI-ASSESSOR	`c4a391d290689932675a…`
11	2008-05-09	First Life Retired	OEM-01	`80e9b0c4d0b721ad9e15…`
12	2008-05-09	Soh Assessment	AI-ASSESSOR	`10bf9338de601870a0d3…`
13	2008-05-09	Second Life Grading	AI-ASSESSOR	`68fef80a07d20458157b…`
14	2008-05-09	Repurposed	REP-01	`4cc3abde64cc485d2820…`
15	2008-05-13	Soh Assessment	AI-ASSESSOR	`93be89d854dcfba6d25d…`
16	2008-05-18	Soh Assessment	AI-ASSESSOR	`95633736ad55e2efc622…`
17	2008-05-23	Soh Assessment	AI-ASSESSOR	`af69bcd2a2e00a0d55c4…`

Head hash (fingerprint of the whole chain)
`af69bcd2a2e00a0d55c4231d89234f356fc9e95102806c7bb01614bae4ee7039`

Each event stores the SHA-256 hash of the event before it and an Ed25519 signature from the actor who issued it. Changing, deleting or re-ordering any event changes the head hash and fails verification. This printed copy is a snapshot; the live record is authoritative.